

1. For charge density

$$\rho(\mathbf{x}) = \frac{e}{4\pi a^2} \frac{e^{-r/a}}{r} \quad (1)$$

the form factor is given by the inverse Fourier transform of (16.31), i.e.,

$$\begin{aligned} f(\mathbf{k}) &= \frac{1}{e} \int d^3x \rho(\mathbf{x}) e^{-i\mathbf{k}\cdot\mathbf{x}} \\ &= \frac{1}{4\pi a^2} \int d\Omega \int_0^\infty r^2 dr \frac{e^{-r/a}}{r} e^{-ikr \cos \theta} \\ &= \frac{1}{4\pi a^2} \cdot 2\pi \int_0^\infty r e^{-r/a} dr \underbrace{\int_0^\pi e^{-ikr \cos \theta} \sin \theta d\theta}_{2 \sin(kr)/kr} \\ &= \frac{1}{a^2 k} \int_0^\infty e^{-r/a} \sin(kr) dr \\ &= \frac{1}{a^2 k} \int_0^\infty e^{-r/a} \left(\frac{e^{ikr} - e^{-ikr}}{2i} \right) dr \\ &= -\frac{1}{a^2 k} \frac{1}{2i} \left(\frac{1}{ik - \frac{1}{a}} + \frac{1}{ik + \frac{1}{a}} \right) \\ &= \frac{1}{1 + k^2 a^2} \end{aligned} \quad (2)$$

2. From (16.33)

$$m = m_0 + \frac{e^2}{3\pi^2 c^2} \int d^3k \frac{|f(\mathbf{k})|^2}{k^2} = m_0 + \frac{e^2}{3\pi^2 c^2} \int d\Omega \underbrace{\int_0^\infty \frac{dk}{(1 + k^2 a^2)^2}}_{\pi/4a} = m_0 + \frac{e^2}{3ac^2} = m_0 + \frac{mc\tau}{2a} \quad (3)$$

3. It is cleaner to use the unsubtracted form (16.32),

$$\begin{aligned} M(\omega) &= m_0 + \frac{e^2}{3\pi^2 c^2} \int \left[\frac{|f(\mathbf{k})|^2}{k^2 - (\omega/c)^2} \right] d^3k && \text{let } q \equiv \frac{\omega}{c} \\ &= m_0 + \frac{e^2}{3\pi^2 c^2} \cdot 4\pi \int_0^\infty \frac{k^2 dk}{(k^2 - q^2)(1 + a^2 k^2)^2} && \text{even integrand} \\ &= m_0 + \frac{e^2}{3\pi^2 c^2} \cdot 4\pi \cdot \frac{1}{2} \underbrace{\int_{-\infty}^\infty \frac{k^2 dk}{(k^2 - q^2)(1 + a^2 k^2)^2}}_I \end{aligned} \quad (4)$$

The integrand has poles at $k = \pm(q + i0^+)$ (simple) and $k = \pm i/a$ (double), and as $|k| \rightarrow \infty$, the modulus of the integrand decays as $1/k^4$, so we can close the contour in either half-plane. Choosing the upper half-plane, where the enclosed poles are $k = q + i0^+$ and $k = i/a$, the integral is given by Cauchy residue theorem

$$I = 2\pi i \left(\text{Res}|_{k=q+i0^+} + \text{Res}|_{k=i/a} \right) \quad (5)$$

where

$$\text{Res}|_{k=q+i0^+} = \frac{q^2}{2q(1 + a^2 q^2)^2} = \frac{q}{2(1 + a^2 q^2)^2} \quad (6)$$

$$\text{Res}|_{k=i/a} = \frac{d}{dk} \left[\frac{k^2}{(k^2 - q^2)a^4(k + i/a)^2} \right]_{k=i/a} = \frac{i(a^2 q^2 - 1)}{4a(1 + a^2 q^2)^2} \quad (7)$$

giving

$$\begin{aligned}
I &= 2\pi i \left[\frac{q}{2(1+a^2q^2)^2} + \frac{i(a^2q^2-1)}{4a(1+a^2q^2)^2} \right] \\
&= 2\pi i \left[\frac{2aq + i(a^2q^2-1)}{4a(1+a^2q^2)^2} \right] \\
&= 2\pi i \left[\frac{-i(1+iaq)^2}{4a(1+iaq)^2(1-iaq)^2} \right] \\
&= \frac{\pi}{2a(1-iaq)^2}
\end{aligned} \tag{8}$$

Hence

$$\begin{aligned}
M(\omega) &= m_0 + \frac{e^2}{3ac^2} \frac{1}{(1-ia\omega/c)^2} = m_0 + \frac{m}{(1-ia\omega/c)^2} \frac{c\tau}{2a} \quad \text{by (3)} \\
&= m \left[1 - \frac{c\tau}{2a} + \frac{\frac{c\tau}{2a}}{(1-ia\omega/c)^2} \right]
\end{aligned} \tag{9}$$

Setting $M(\omega) = 0$ and solving for ω , we have

$$\left(1 - \frac{ia\omega}{c}\right)^2 = \frac{1}{1 - \frac{2a}{c\tau}} \quad \Rightarrow \quad \omega = -\frac{ic}{a} \left(1 \pm \frac{1}{\sqrt{1 - \frac{2a}{c\tau}}}\right) \tag{10}$$

4. From (3), the condition $m_0 \leq (\geq) 0$ is equivalent to $a \leq (\geq) c\tau/2$.

When $m_0 < 0$, (10) indicates that the zero ω is purely imaginary, with + solution in the lower half-plane and - solution in the upper half-plane, i.e., a runaway solution.

On the other hand, when $m_0 > 0$, ω has both real and imaginary parts, and the imaginary part is always negative, no runaway solution.

For limiting forms,

- When $a/c\tau \ll 1$, we have

$$\omega_{\pm} \approx -\frac{ic}{a} \left[1 \pm \left(1 + \frac{a}{c\tau}\right)\right] \quad \Rightarrow \quad \omega_{+} \approx -\frac{2ic}{a} \quad \omega_{-} \approx \frac{i}{\tau} \tag{11}$$

where ω_{-} reproduces the point-charge Abraham-Lorentz runaway $\omega = i/\tau$ of (16.35), and the + root runs off to $-i\infty$.

- For $a/c\tau \gg 1$,

$$\omega_{\pm} \approx -\frac{ic}{a} \mp \frac{c}{a} \sqrt{\frac{c\tau}{2a}} \tag{12}$$

both of which approach the origin from below.